

4. b

Let son's age be x years. Then, father's age = $(3x)$ years.

Five years ago, father's age = $(3x - 5)$ years and son's age = $(x - 5)$ years.

So, $3x - 5 = 4(x - 5) \Rightarrow 3x - 5 = 4x - 20 \Rightarrow x = 15$

5. d

Let number of horses = number of men = x .

Then, number of legs = $4x + 2 \times (x/2) = 5x$.

So, $5x = 70$ or $x = 14$

6. b

Let the number of cows be x and the number of hens be y .

Then, $4x + 2y = 2(x + y) + 14 \Rightarrow 4x + 2y = 2x + 2y + 14 \Rightarrow 2x = 14 \Rightarrow x = 7$

7. c

Girls = $3/5$, Boys = $(1 - \frac{3}{5}) = \frac{2}{5}$

Fraction of students absent = $\frac{2}{9}$ of $\frac{3}{5} + \frac{1}{4}$ of $\frac{2}{5} =$

$$\frac{6}{45} + \frac{1}{10} = \frac{21}{90} = \frac{7}{30}$$

Therefore, Fraction of students present =

$$(1 - \frac{7}{30}) = \frac{23}{30}$$

8. b

Let the number of 20-paise coins be x . Then, number of 25-paise coins = $(324 - x)$.

Therefore $0.20x + 0.25(324 - x) = 71 \Rightarrow 20x + 25(324 - x) = 7100$

$\Rightarrow 5x = 1000 \Rightarrow x = 200$. Hence, number of 25-paise coins = $(324 - x) = 124$

9. c

Let the total number of sweets be $(25x + 8)$.

Then, $(25x + 8) - 22$ is divisible by 28

$\Rightarrow (25x - 14)$ is divisible by 28 $\Rightarrow 28x - (3x + 14)$ is divisible by 28

$\Rightarrow (3x + 14)$ is divisible by 28 $\Rightarrow x = 14$.

Therefore Total number of sweets = $(25 \times 14 + 8) = 358$

10. a

Let the ten's digit be x . Then, unit's digit = $(11 - x)$.

So, number = $10x + (11 - x) = 9x + 11$.

Therefore $(9x + 11) + 27 = 10(11 - x) + x \Rightarrow 9x + 38 = 110 - 9x \Rightarrow 18x = 72 \Rightarrow x = 4$.

Thus, ten's digit = 4 and unit's digit = 7.
Hence, required number = 47

11. a

Required sum = $(80 - 3 \times 3)$ years = $(80 - 9)$ years = 71 years

12. c

Let there be $(x + 1)$ members, then,

father's share = $\frac{1}{4}$, share of each other member = $\frac{3}{4x}$

Therefore, $3(\frac{3}{4x}) = \frac{1}{4} \Rightarrow 4x = 36 \Rightarrow x = 9$

Hence, total number of family members = 10

13. d

'All but eleven died' means 'All except eleven died' i.e. 11 sheep remained alive

14. a

There were all sparrows but six' means that six birds were not sparrows but only pigeons and ducks.

Similarly, number of sparrows + number of ducks = 6 and number of sparrows + number of pigeons = 6.

This is possible when there are 3 sparrows, 3 pigeons and 3 ducks i.e. 9 birds in all

15. a

Clearly, the smallest such number is 3.

D
[D]
[D]
D

Three ducks can be arranged as shown above to satisfy all the three given conditions.

16. d

Let the number of adults and children be $2x$ and $3x$ respectively.

Then, literate population = $(100 - 40)\%$ of $2x + 85\%$ of $3x$

$$= \left(\frac{60}{100} \times 2x\right) + \left(\frac{85}{100} \times 3x\right) = \frac{6x}{5} + \frac{51x}{20} = \frac{75x}{20}$$

Therefore, required percentage = $\left(\frac{75x}{20} \times \frac{1}{5x} \times 100\right)\% = 75\%$

17. b

Let the number of boys be x . Then, $(3/4)x = 18$ or $x = 18 \times (4/3) = 24$.

If total number of students is y , then $(2/3)y = 24$ or $y = 24 \times (3/2) = 36$.

Therefore Number of girls in the class = $(36 - 24) = 12$

18. d

Let salary = Rs. x . Then tips = Rs. $(\frac{5}{4}x)$

Total income = Rs. $(x + \frac{5}{4}x) = \text{Rs. } (\frac{9x}{4})$

Therefore, required fraction = $(\frac{5x}{4} \times \frac{4}{9x}) = \frac{5}{9}$

19. c

Clearly, from 1 to 100, there are ten numbers with 3 as the unit's digit- 3, 13, 23, 33, 43, 53, 63, 73, 83, 93; and ten numbers with 3 as the ten's digit - 30, 31, 32, 33, 34, 35, 36, 37, 38, 39.

So, required number = $10 + 10 = 20$

20. d

Less cats, more days (Indirect proportion)

Less mice, less days (Direct proportion)

Let the required number of days be x .

Cat 25 : 100 } $\therefore x : 100$

Mice 100 : 25 }

Therefore, $(\frac{25 \times 100 \times 100}{100 \times 25}) = 100$

21. b

Manick's present age = 12 years, Rahul's present age = 4 years.

Let Manick be twice as old as Rahul after x years from now.

Then, $12 + x = 2(4 + x)$ $12 + x = 8 + 2x$ $x = 4$.

Hence, Manick's required age = $12 + x = 16$ years.

22. c

Let the number of pineapples & watermelons be x & y respectively.

Then $7x + 5y = 38$

$$5y = 38 - 7x$$

$$y = \frac{38 - 7x}{5}$$

Clearly y is a whole number, only when $(38 - 7x)$ is divisible by 5

This happens when x is 4

23. b

Angle traced by hour hand in $13/3$ hours =

$$(\frac{360}{12} \times \frac{13}{3})^\circ = 130^\circ$$

Angle traced by min. hand in 20 min =

$$(\frac{360}{60} \times 20)^\circ = 120^\circ$$

Required angle = $(130 - 120)^\circ = 10^\circ$

24. c

In 12 hours, they are at right angles 22 times

Therefore, In 24 hours, they are at right angles 44 times

25. b

Angle traced by hour hand in $17/2$ hours =

$$(\frac{360}{12} \times \frac{17}{2})^\circ = 255^\circ$$

Angle traced by min. hand in 30 min =

$$(\frac{360}{60} \times 30)^\circ = 180^\circ$$

Required angle = $(255 - 180)^\circ = 75^\circ$

26. b

The hands of a clock point in opposite directions (in the same straight line) 11 times in every 12 hours. (Because between 5 and 7 they point in opposite directions at 6 o'clock only).

So, in a day, the hands point in the opposite directions 22 times

27. b

Here initially 12 birds were there when first half i.e. 6 killed, left is 6 and again half i.e. 3 killed, only three left in the end

28. d

360° in 12 hours. This means that for every hour, we have the hour hand move $360/12$, or 30°

Therefore, in 9 hours = $9 \times 30^\circ = 270^\circ$

29. b

$$\text{Time} = \left(\frac{100 \times 81}{450 \times 4.5} \right) \text{ years} \\ = 4 \text{ years}$$

30. a

Let the average age of the whole team by x years.

$$\text{Therefore, } 11x - (26 + 29) = 9(x - 1)$$

$$\Rightarrow 11x - 9x = 46$$

$$\Rightarrow 2x = 46$$

$$\Rightarrow x = 23.$$

So, average age of the team is 23 years

31. d

Let A, B, C represent their respective weights.

Then, we have:

$$A + B + C = (45 \times 3) = 135 \dots (i)$$

$$A + B = (40 \times 2) = 80 \dots (ii)$$

$$B + C = (43 \times 2) = 86 \dots (iii)$$

Adding (ii) and (iii), we get: $A + 2B + C = 166 \dots$

(iv)

Subtracting (i) from (iv), we get: $B = 31$.

Therefore B's weight = 31 kg

32. c

Therefore, required age

$$= \left(\frac{50.25 \times 16 + 45.15 \times 8}{16 + 8} \right)$$

$$= \left(\frac{804 + 361.20}{24} \right) = 48.55$$

33. c

Let there be x pupils in the class

$$\text{Total increase in marks} = \left(x \times \frac{1}{2} \right) = \frac{x}{2}$$

$$\text{Therefore, } \frac{x}{2} = [83 - 63] \text{ which implies, } x = 40$$

34. a

$$\text{Total number of balls} = (2 + 3 + 2) = 7$$

Let S be the sample space.

Then, $n(S)$ = Number of ways of drawing 2 balls out of 7

$$= {}^7C_2$$

$$= \frac{(7 \times 6)}{(2 \times 1)} = 21$$

Let E = Event of drawing 2 balls, none of which is blue.

35. c

$$\text{In two throws of a dice, } n(S) = (6 \times 6) = 36$$

Let E = event of getting a sum = $\{(3, 6), (4, 5), (5, 4), (6, 3)\}$

Therefore,

$$P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

36. d

$$\text{Quantity of milk} = \left(60 \times \frac{2}{3} \right) \text{ litres} = 40 \text{ litres}$$

$$\text{Quantity of water in it} = (60 - 40) \text{ litres} = 20 \text{ litres}$$

$$\text{New ratio} = 1 : 2$$

Let quantity of water to be added further be x litres

$$\text{Then, milk : water} = \left(\frac{40}{20+x} \right)$$

$$\text{Now, } \left(\frac{40}{20+x} \right) = \frac{1}{2}$$

$$\Rightarrow 20 + x = 80 \Rightarrow x = 60$$

Therefore, quantity of water to be added = 60 litres

37. b

Let the number be $3x$ and $5x$

$$\text{Then, } \frac{3x-9}{5x-9} = \frac{12}{23}$$

$$\Rightarrow 23(3x - 9) = 12(5x - 9)$$

$$\Rightarrow 9x = 99 \Rightarrow x = 11$$

Therefore, the smaller number = $(3 \times 11) = 33$

38. c

On 31st December, 2005 it was Saturday.

Number of odd days from the year 2006 to the year 2009 = $(1 + 1 + 2 + 1) = 5$ days.

Therefore, on 31st December 2009, it was Thursday.

Thus, on 1st Jan, 2010 it is Friday

39. c

The year 2004 is a leap year. It has 2 odd days

Therefore, the day on 8th Feb, 2004 is 2 days before the day on 8th Feb, 2005

Hence, this day is Sunday

40. d

28 May, 2006 = (2005 years + Period from 1.1.2006 to 28.5.2006)

Odd days in 1600 years = 0

Odd days in 400 years = 0

5 years = (4 ordinary years + 1 leap year) = $(4 \times 1 + 1 \times 2) \equiv 6$ odd days

Jan.	Feb.	March	April	May
(31 + 28	+ 31	+ 30	+ 28)	=
148 days				

Therefore, 148 days = (21 weeks + 1 day) $\equiv 1$

odd day.

$$\text{Total number of odd days} = (0 + 0 + 6 + 1) = 7$$

$\equiv 0$ odd day.

Given day is Sunday.

41. d

Here initially 18 birds were there when first half and one killed i.e. $9 + 1$ killed, left is 8 and again half and one killed i.e. 5 killed, only three left in the end

42. a

$$132 = 4 \times 3 \times 11$$

So, if the number divisible by all the three number 4, 3 and 11, then the number is divisible by 132 also.

$$264 \Rightarrow 11, 3, 4 (/)$$

$$396 \Rightarrow 11, 3, 4 (/)$$

$$462 \Rightarrow 11, 3 (X)$$

$$792 \Rightarrow 11, 3, 4 (/)$$

$$968 \Rightarrow 11, 4 (X)$$

$$2178 \Rightarrow 11, 3 (X)$$

$$5184 \Rightarrow 3, 4 (X)$$

$$6336 \Rightarrow 11, 3, 4 (/)$$

Therefore the following numbers are divisible by 132 : 264, 396, 792 and 6336.

Required number of number = 4

43. b

Formula: (Divisor * Quotient) + Remainder = Dividend.

$$(56 * Q) + 29 = D \dots\dots\dots (1)$$

$$D \% 8 = R \dots\dots\dots (2)$$

From equation (2),

$$((56 * Q) + 29) \% 8 = R.$$

$$\Rightarrow \text{Assume } Q = 1.$$

$$\Rightarrow (56 + 29) \% 8 = R.$$

$$\Rightarrow 85 \% 8 = R$$

$$\Rightarrow 5 = R.$$

44. a

$$\text{Given expression} = \frac{(12)^3 \times 6^4}{432}$$

$$= \frac{(12)^3 \times 6^4}{12 \times 6^2} = (12)^2 \times 6^2$$

$$= 72^2 = 5184$$

45. d

$$(4 + 5 + 2) - (1 + 6 + 3) = 1, \text{ not divisible by 11.}$$

$$(2 + 6 + 4) - (4 + 5 + 2) = 1, \text{ not divisible by 11}$$

46. b

Clearly, the required number would be such that it leaves a remainder of 1 when divided by 3, 4, 5 or 6 and no remainder when divided by 7. Such a number is 301.

47. d

Let Sunita's present age be x years.

Then, Reena's present age is $2x$ years.

Three years ago, Sunita's age = $(x - 3)$ and

Reena's age = $(2x - 3)$.

So, $(2x - 3) - 3(x - 3)$ or $2x - 3 - 3x + 9$ or $x - 6$.

Reena's present age = $2x = 12$ years.

48. a

The total age of the two players which were replaced is the same as that of the total age of two players which replaced them. Therefore, the new average will remain unchanged.

49. a

The age of Ram = 8 years. The age of Geeta = $8 - 2 = 6$ years.

Therefore, the age of Kamal = $6 \times 5 = 30$ years

50. c

The age of mother = Father's age - 5 = $45 - 5 = 40$ years

$$\text{The age of son} = \frac{40}{2} - 10 = \frac{40 - 20}{2} = 10 \text{ years}$$

51. c

Suppose the age of Johnny is x years and that of Rahul is y years.

According to question Age of Hari = $2x = 5y$ or, $2x - 5y = 0 \dots (i)$

Again $x = 3 + y$ or, $x - y = 3 \dots (ii)$

On solving equations (i) and (ii),

we get $y = 2$

Therefore Age of Johnny = $3 + y = 3 + 2 = 5$ years

52. b

19 Total age of boys = $19 \times 21 = 399$ years

Total age of 19 boys and teacher = $20 \times 22 = 440$ years \

Therefore, Teacher's age = $440 - 399 = 41$ years

53. c

Age of C = $(3 \times 20) - (2 \times 19) = 60 - 38 = 22$ years

54. c

Suppose the present age of daughter = x years
and, the present age of mother = y years
According to questions, $3(x - 5) = (y - 5)$

$$3x - 15 = y - 5$$

$$3x - y = 10 \text{(i)}$$

$$2(x + 5) = y + 5$$

$$2x + 10 = y + 5$$

$$2x - y = -5 \text{(ii)}$$

From equations (i) and (ii)

$$3x - 2x = 10 - (-5)$$

$$x = 15 \text{ years}$$

55. b
Weight of new person (using formula) = wt. of
replaced person + $n \times$ increase in
Average = $60 + 8 \times 1 = 68$ kg

56. d
Suppose the present age of Leela is x years.
Therefore, the present age of Neela = $3x$ years
10 years ago $5(x - 10) = (3x - 10)$
 $5x - 50 = 3x - 10$
 $5x - 3x = 50 - 10$
 $2x = 40$
 $x = 40/2 = 20$ years

57. c
Amit = 17 years Rakesh = $17 - 5 = 12$ years
Therefore Anil = $12 + 3 = 15$ years

58. d
Age of son = 6 years
Age of father = $6 \times 5 = 30$ years
Let after x years father will be 4 times of son.
 $30 + x = 4(6 + x)$
 $4x - x = 30 - 24$
 $3x = 6$
 $x = 2$ years

59. b
Suppose the age of son is x years
Therefore, age of father = $10x$ years
According to question
 $\frac{10x+x}{2} = 22$
 $11x = 44$
Therefore $x = 4$
Age of father = $10 \times 4 = 40$ years

60. a
Suppose the age of Shyam = x years
Age of Ram = $2x$ years
Age of Suresh = $4x$ years

According to question $x + 2x + 4x = 70$ or, $7x = 70$
Therefore $x = 10$
Age of Ram = $2x = 2 \times 10 = 20$ years

61. b
Age of Mani = 60 years
Age of Prabhu = $60/2 = 30$ years
Age of Ramona = $30/2 = 15$ years

62. d
Suppose, the age of Nikhil is 1 year,
therefore, age of Rohan = $1 + 8 = 9$ years
After 7 years,
Nikhil will be 8 years old and Rohan will be 16 years old

63. c
50 men can dig 40 holes in 30 days
Therefore 25 men can dig 20 holes in $\frac{50 \times 20 \times 40}{23 \times 30}$
 $= 30$ days

64. b
4 cats can kill 4 rats in 4 minutes. So, 8 cats can
kill 8 rats in 4 minutes

65. c
Man plants 10 trees in an hour. He worked for
10 hours, but took rest for 30 minutes after
every hour. It means he worked for two hours
in every three hours. Thus, during 9 hours he
worked for 6 hours. In six hours he planted $6 \times 10 = 60$ trees. In the tenth hour, he planted
another 10 trees. Therefore, total number of
trees = $60 + 10 = 70$

66. d
Total number of persons in the family = $2 + 3 + 3 = 8$

67. c
Number of heads = $B + D = 7$...(i)
Number of legs = $2B + 4D = 20$ or $B + 2D = 10$
...(ii)
From statements (i) and (ii) $B = 4$ and $D = 3$

68. b
25 feet 6 inches = $25 \times 12 + 6 = 300 + 6 = 306$
inches
 $306/3 = 102$ inches
Therefore 102 inches = 8 feet 6 inches

69. c

Required number of cuttings = AR^{n-1}
Here, $A = 2$, $R = 2$ and $n = 10$
 $2 \times 2^{10-1} = 2 \times 2^9 = 1024$

70. c
From 1 to 10 = 1
From 11 to 20 = 1
From 21 to 30 = 1
i.e., From 1 to 80 = 9
From 81 to 90 = 10
From 91 to 99 = 1
Total number of 8s = 20

71. a
Such numbers are : 24, 42, 45, 48, 54, 84

72. c
17, 34, 51, 68, 85, 102, 119, 136, 153
In these range of numbers, 68 and 136 are
divisible by 4.

68/4 gives you 17.
136/4 gives you 34.
17, 34, 51, 85, 102, 119, 153.
So, that will be 7 numbers

73. b

$$\frac{0.1}{0.01} = 10$$

74. b
Shadow of one feet high pole = $20/18$
Shadow of 27 feet high pole = $\frac{20}{18} \times 27 = 30$
feet

75. c
According to question 16 sheep died.
Therefore, 18 sheep were left

76. d
25 students play hockey. 30 students play
football. 8 students play neither game.
The students who play hockey, football and
both the games = $50 - 8 = 42$
The number of students who play both hockey
and football = $(25 + 30) - 42 = 55 - 42 = 13$

77. c
From question Number of girls = 40% i.e 300
Number of boys = 60%
 $\frac{300}{40} \times 60 = 450$

78. d
40% of $1620 + 30\%$ of $960 = x\%$ of 5200

$$\frac{40}{100} \times 1620 + \frac{30}{100} \times 960 = \frac{x}{100} \times 5200$$

Solving above eqn we get $x = 18$

79. c
Girls Boys = 90/100
So, the number of girls in the town is 90

80. d
Suppose basic pay of A = Rs. x
Basic pay of B = Rs. y
Ratio between basic pays
 $\frac{x \times 165}{100} = \frac{y \times 180}{100}$

$$\frac{x}{y} = \frac{180}{165} = 12:11$$

81. b
6 litre of 40% alcohol contains $\frac{60 \times 6}{100} = 3.6$ litre
water
According to the question, 8 litre of solution
also contains 3.6 litre water
Therefore % of water in the solution = $\frac{3.6}{8} \times 100 = 45\%$

82. a
Suppose Ravi has Rs. x
Rama has Rs. $x + \frac{x}{3}$
Therefore $x + \frac{x}{3} - x = 4$
Therefore $x = 12$
Rama has $12 + \frac{12}{3} =$ Rs. 16

83. d
According to question
 $x + \frac{x}{2} + \frac{x}{4} + 1 = 71$
Solving the above equation we get
 $X = 40$

84. d
10 boys walk 10 km in 10 days, i.e., any
number of boys can walk 10 km in 10 days

85. a
 $Speed = \frac{Distance}{Time} = \frac{150m}{12sec}$
 $\frac{150}{12} \times \frac{18}{5} = 45$ kmph